

IoT Network Intrusion Detection System

Paper Replication Matrix Guide (12 - 11 - 11 - 11 Division)

PROJECT SCOPE: Computer Networks Semester Project	TOTAL PAPER MATRIX: 45 Runs (Strict Paper Replication)
BASE PAPER: Wang et al. (Journal of Supercomputing, 2024)	GITHUB REPOSITORY: Harish8955/IoT_NIDS_Project

Group Member	Assigned Paper Dataset & Model Runs	Assigned Runs
Harish Srinivas	UNSW-NB15 Proposed Model & Baseline Runs (50, 100, 200 Epochs)	12 Runs
Edwin Sundaraj	CSE-CIC-IDS2018 Proposed, 1D-CNN & ResNet Runs + UNSW Baselines	11 Runs
Anish Thondepu	CIC-IOT2023 Proposed, 1D-CNN & ResNet Runs + UNSW Baselines	11 Runs
Lakshan	CSE-CIC-IDS2018 & CIC-IOT2023 ResNeSt and ResNet-BiGRU Baselines	11 Runs

1. HARISH SRINIVAS — ASSIGNED PAPER RUNS (12 RUNS)

```
python train.py --dataset data/UNSW_NB15_combined.csv --dataset_name UNSW-NB15 --target_col attack_cat --model proposed --epochs 50 --balance
python train.py --dataset data/UNSW_NB15_combined.csv --dataset_name UNSW-NB15 --target_col attack_cat --model proposed --epochs 100 --balance
python train.py --dataset data/UNSW_NB15_combined.csv --dataset_name UNSW-NB15 --target_col attack_cat --model proposed --epochs 200 --balance
python train.py --dataset data/UNSW_NB15_combined.csv --dataset_name UNSW-NB15 --target_col attack_cat --model cnn --epochs 50 --balance
python train.py --dataset data/UNSW_NB15_combined.csv --dataset_name UNSW-NB15 --target_col attack_cat --model cnn --epochs 100 --balance
python train.py --dataset data/UNSW_NB15_combined.csv --dataset_name UNSW-NB15 --target_col attack_cat --model cnn --epochs 200 --balance
python train.py --dataset data/UNSW_NB15_combined.csv --dataset_name UNSW-NB15 --target_col attack_cat --model resnet --epochs 50 --balance
python train.py --dataset data/UNSW_NB15_combined.csv --dataset_name UNSW-NB15 --target_col attack_cat --model resnet --epochs 100 --balance
python train.py --dataset data/UNSW_NB15_combined.csv --dataset_name UNSW-NB15 --target_col attack_cat --model resnet --epochs 200 --balance
python train.py --dataset data/UNSW_NB15_combined.csv --dataset_name UNSW-NB15 --target_col attack_cat --model resnet --epochs 200 --balance
python train.py --dataset data/UNSW_NB15_combined.csv --dataset_name UNSW-NB15 --target_col attack_cat --model resnet --epochs 100 --balance
python train.py --dataset data/UNSW_NB15_combined.csv --dataset_name UNSW-NB15 --target_col attack_cat --model resnet --epochs 200 --balance
```

2. EDWIN SUNDARAJ — ASSIGNED PAPER RUNS (11 RUNS)

```
python train.py --dataset data/CIC_IDS2018_combined.csv --dataset_name CIC-IDS2018 --target_col Label --model proposed --epochs 50 --balance
python train.py --dataset data/CIC_IDS2018_combined.csv --dataset_name CIC-IDS2018 --target_col Label --model proposed --epochs 100 --balance
python train.py --dataset data/CIC_IDS2018_combined.csv --dataset_name CIC-IDS2018 --target_col Label --model proposed --epochs 200 --balance
python train.py --dataset data/CIC_IDS2018_combined.csv --dataset_name CIC-IDS2018 --target_col Label --model cnn --epochs 50 --balance
python train.py --dataset data/CIC_IDS2018_combined.csv --dataset_name CIC-IDS2018 --target_col Label --model cnn --epochs 100 --balance
python train.py --dataset data/CIC_IDS2018_combined.csv --dataset_name CIC-IDS2018 --target_col Label --model cnn --epochs 200 --balance
python train.py --dataset data/CIC_IDS2018_combined.csv --dataset_name CIC-IDS2018 --target_col Label --model resnet --epochs 50 --balance
python train.py --dataset data/CIC_IDS2018_combined.csv --dataset_name CIC-IDS2018 --target_col Label --model resnet --epochs 100 --balance
python train.py --dataset data/CIC_IDS2018_combined.csv --dataset_name CIC-IDS2018 --target_col Label --model resnet --epochs 200 --balance
python train.py --dataset data/UNSW_NB15_combined.csv --dataset_name UNSW-NB15 --target_col attack_cat --model resnet --epochs 50 --balance
python train.py --dataset data/UNSW_NB15_combined.csv --dataset_name UNSW-NB15 --target_col attack_cat --model resnet --epochs 100 --balance
```

3. ANISH THONDEPU — ASSIGNED PAPER RUNS (11 RUNS)

```
python train.py --dataset data/CICIOT23_combined.csv --dataset_name CIC-IOT2023 --target_col label --model proposed --epochs 50 --balance
python train.py --dataset data/CICIOT23_combined.csv --dataset_name CIC-IOT2023 --target_col label --model proposed --epochs 100 --balance
python train.py --dataset data/CICIOT23_combined.csv --dataset_name CIC-IOT2023 --target_col label --model proposed --epochs 200 --balance
python train.py --dataset data/CICIOT23_combined.csv --dataset_name CIC-IOT2023 --target_col label --model cnn --epochs 50 --balance
python train.py --dataset data/CICIOT23_combined.csv --dataset_name CIC-IOT2023 --target_col label --model cnn --epochs 100 --balance
python train.py --dataset data/CICIOT23_combined.csv --dataset_name CIC-IOT2023 --target_col label --model cnn --epochs 200 --balance
python train.py --dataset data/CICIOT23_combined.csv --dataset_name CIC-IOT2023 --target_col label --model resnet --epochs 50 --balance
python train.py --dataset data/CICIOT23_combined.csv --dataset_name CIC-IOT2023 --target_col label --model resnet --epochs 100 --balance
python train.py --dataset data/CICIOT23_combined.csv --dataset_name CIC-IOT2023 --target_col label --model resnet --epochs 200 --balance
python train.py --dataset data/UNSW_NB15_combined.csv --dataset_name UNSW-NB15 --target_col attack_cat --model resnet --epochs 50 --balance
python train.py --dataset data/UNSW_NB15_combined.csv --dataset_name UNSW-NB15 --target_col attack_cat --model resnet --epochs 100 --balance
```

4. LAKSHAN — ASSIGNED PAPER RUNS (11 RUNS)

```
python train.py --dataset data/CIC_IDS2018_combined.csv --dataset_name CIC-IDS2018 --target_col Label --model resnest --epochs 50 --balance
python train.py --dataset data/CIC_IDS2018_combined.csv --dataset_name CIC-IDS2018 --target_col Label --model resnest --epochs 100 --balance
python train.py --dataset data/CIC_IDS2018_combined.csv --dataset_name CIC-IDS2018 --target_col Label --model resnest --epochs 200 --balance
python train.py --dataset data/CIC_IDS2018_combined.csv --dataset_name CIC-IDS2018 --target_col Label --model resnet-gru --epochs 50 --balance
python train.py --dataset data/CIC_IDS2018_combined.csv --dataset_name CIC-IDS2018 --target_col Label --model resnet-gru --epochs 100 --balance
python train.py --dataset data/CIC_IDS2018_combined.csv --dataset_name CIC-IDS2018 --target_col Label --model resnet-gru --epochs 200 --balance
python train.py --dataset data/CIC_IOT23_combined.csv --dataset_name CIC-IOT2023 --target_col label --model resnest --epochs 50 --balance
python train.py --dataset data/CIC_IOT23_combined.csv --dataset_name CIC-IOT2023 --target_col label --model resnest --epochs 100 --balance
python train.py --dataset data/CIC_IOT23_combined.csv --dataset_name CIC-IOT2023 --target_col label --model resnest --epochs 200 --balance
python train.py --dataset data/CIC_IOT23_combined.csv --dataset_name CIC-IOT2023 --target_col label --model resnet-gru --epochs 50 --balance
python train.py --dataset data/CIC_IOT23_combined.csv --dataset_name CIC-IOT2023 --target_col label --model resnet-gru --epochs 100 --balance
```

5. Syncing Results Back to Master GitHub Repository

When training completes, each group member syncs their generated results back to the master Excel summary table (results/experiment_summary.csv):

```
git pull
git add results/
git commit -m "Added experiment training results"
git push
```